例题与习题（一）

1、功的互等定理

2、单位载荷法应用

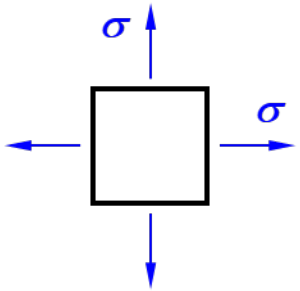
3、温度变形导致梁弯曲

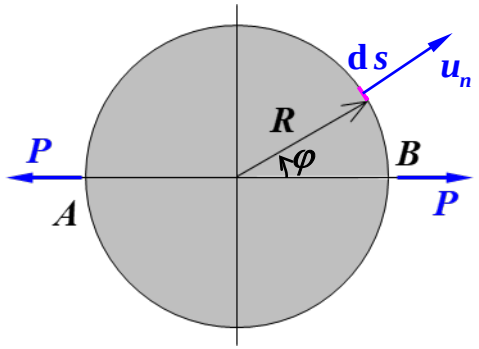
4、分段冲击问题

功的互等定理

例 图示匀质等厚圆板，半径R，厚度b，弹性模量E，泊松比 μ，沿直径受一对力P作用。求面积变化ΔA.

解：
$$\Delta A = \oint u_n \, \mathrm{d}s$$

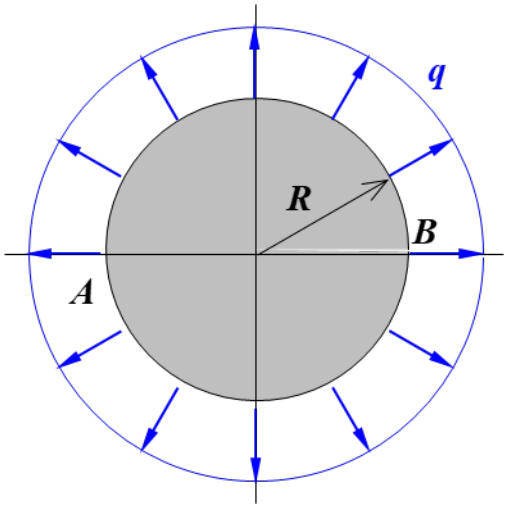




构造均布法向载荷

$$\boxed{P\Delta D_{AB} = \oint u_n q \, \mathrm{d}s}$$

轴对称受力，二向等拉



$$\sigma = \frac{q}{b}, \quad \varepsilon = \frac{1-\mu}{E}\sigma \qquad \Delta D_{AB} = 2R\varepsilon$$

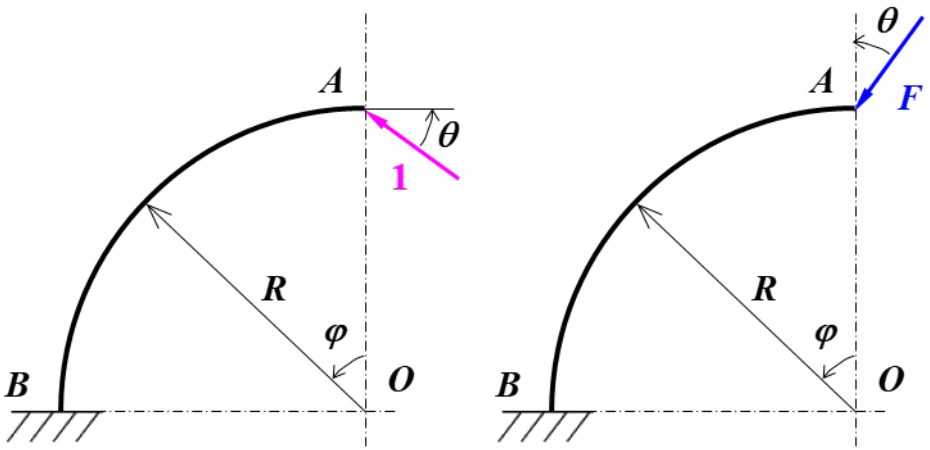
$$\Delta A = \frac{2(1-\mu)PR}{Eb}$$

单位载荷法应用

例 图示1/4圆弧小曲率杆*AB*与*CD*相同，*EI*为常数，半径为*R*，自由端受集中力*F*作用。为使A截面的总位移沿*F*方向，θ=?

$$M(\varphi)=F\cos\theta R\sin\varphi$$
$$-F\sin\theta R(1-\cos\varphi)$$

$$\overline{M}(\varphi)=-\sin\theta R\sin\varphi$$
$$-\cos\theta R(1-\cos\varphi)$$

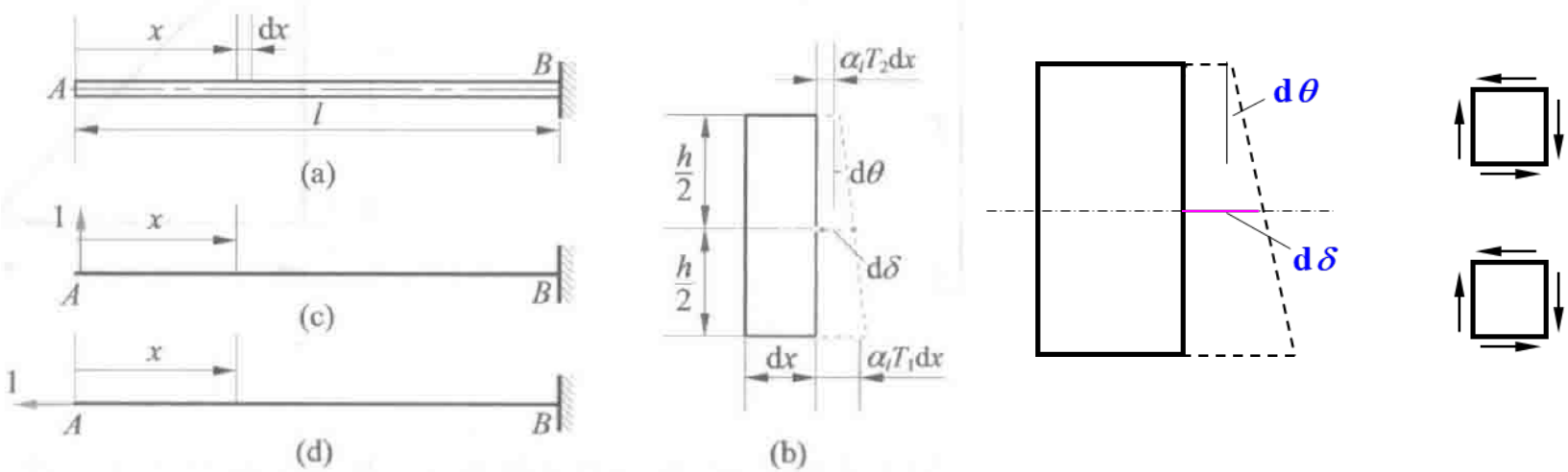


$$\Delta_{AF}=\int_l\frac{\overline{M}(\varphi)M(\varphi)}{EI}R\,\mathrm{d}\varphi=0$$

$$\tan 2\theta=\frac{2}{\pi-4}\qquad\theta=-66.7^\circ$$

梁为什么会弯

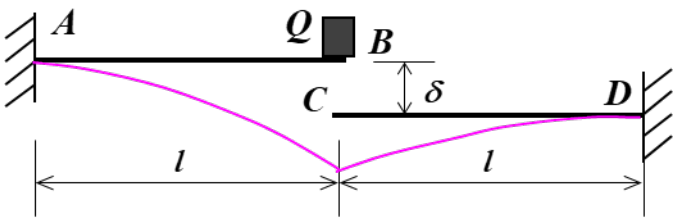
习题13-31 温度导致纵向变形不一致



- 温度变化 → 纵向线性伸长 → 截面转动 → 切应变；
- 维持该切应变的切应力不平衡 → 无切应变；
- 轴线变弯 → 截面与轴线正交。
- 若温度变化上下对称，是否变弯？

分段冲击问题

例 图示悬臂梁AB与CD相同，EI为常数，两自由端B与C间距为 $\delta = Ql^3/3EI$，重物Q突然施加于B处，求：C截面的挠度f_C.



对于AB梁

$$K_d = 1 + \sqrt{1 + \frac{2h}{\Delta_{st}}} = 2$$

$$\Delta_d = 2\frac{Ql^3}{3EI} > \delta$$

二梁接触

$$Q(\delta + f_C) = \frac{k_{AB}}{2}(\delta + f_C)^2 + \frac{k_{CD}}{2} f_C^{\,2}$$

- 绕过冲击载荷
- 以位移表示变形能
- 能量守恒

$$k_{AB} = k_{CD} = \frac{3EI}{l^3}$$

$$f_C = \frac{\sqrt{2}}{2}\delta$$

$$V_\varepsilon = \frac{k\Delta^2}{2} = \int_l \frac{M^2(x)}{2EI}\mathrm{d}x = \frac{F_d\Delta_d}{2} = \frac{F_d^2}{2k}$$

例题与习题（二）

1、单位载荷法应用

2、对称性应用

3、选好基本系统

4、静不定结构冲击

5、卡氏定理求广义位移

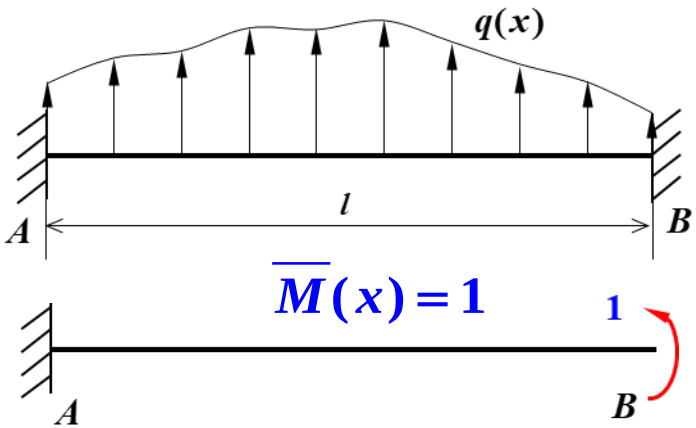
6、反对称内力与位移求解

单位载荷法应用

例 图示两端固定梁，EI为常数，

证明 $\int_l M(x)\mathrm{d}x = 0$

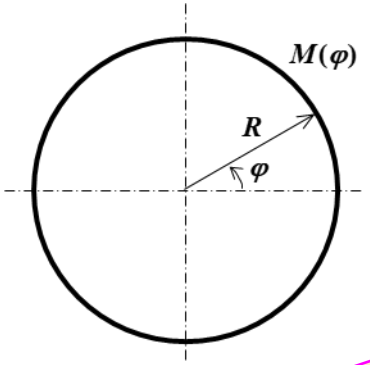
$$\theta_B = \int_l \frac{\overline{M}(x)M(x)}{EI}\mathrm{d}x = 0$$

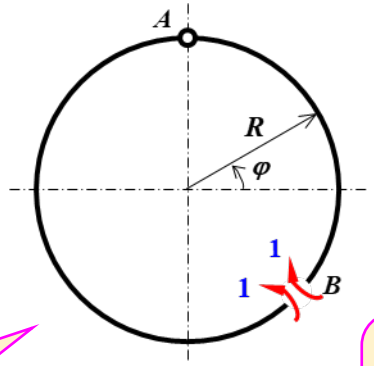


讨论：圆环受任意载荷

1）封闭圆环，EI为常数，

可证 $\oint M(\varphi)\mathrm{d}\varphi = 0$





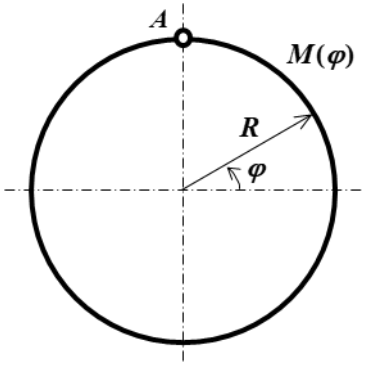
是基本系统吗？

不平衡

2）圆环含铰链

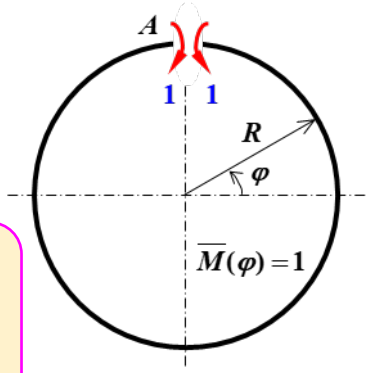
$$\theta_{A^-/A^+} = \oint \overline{M}(\varphi)\mathrm{d}\theta$$

$$\oint \mathrm{d}\theta = \frac{R}{EI}\oint M(\varphi)\mathrm{d}\varphi \neq 0$$



3）任意B处切开

$$\theta_{B^-/B^+} = 0$$

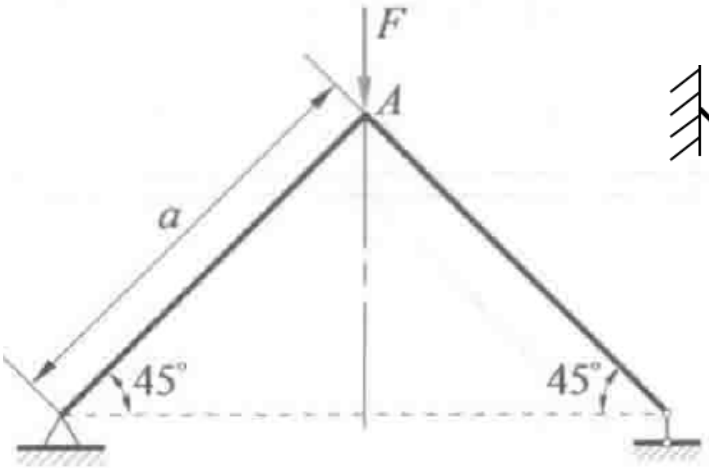


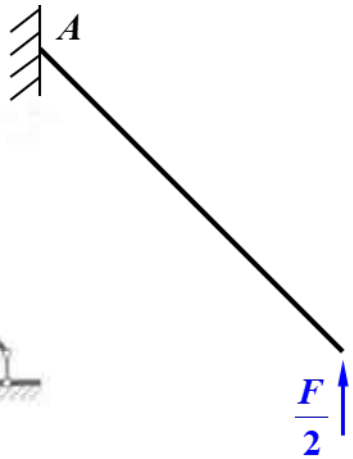
- 基本系统是静定基
- 单位载荷 → 静力许可场

对称性的应用

习题13-21 图示刚架，

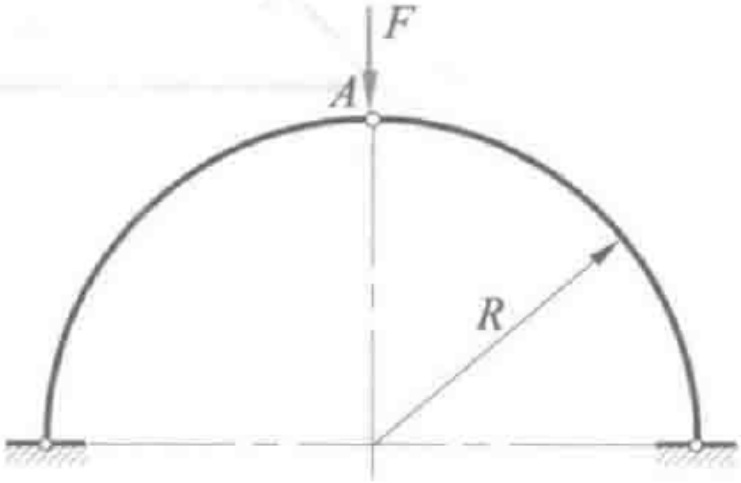
求：f_A

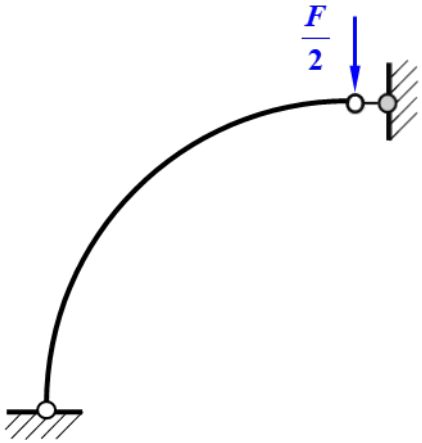


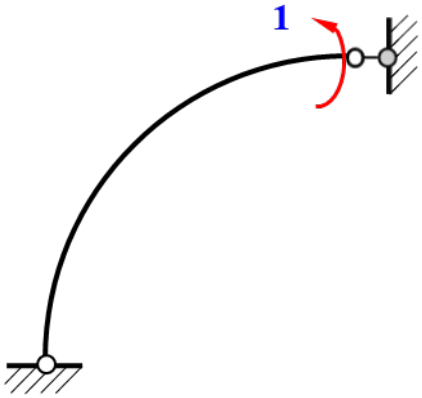


习题13-26 图示结构，

求：铰链两侧相对转角



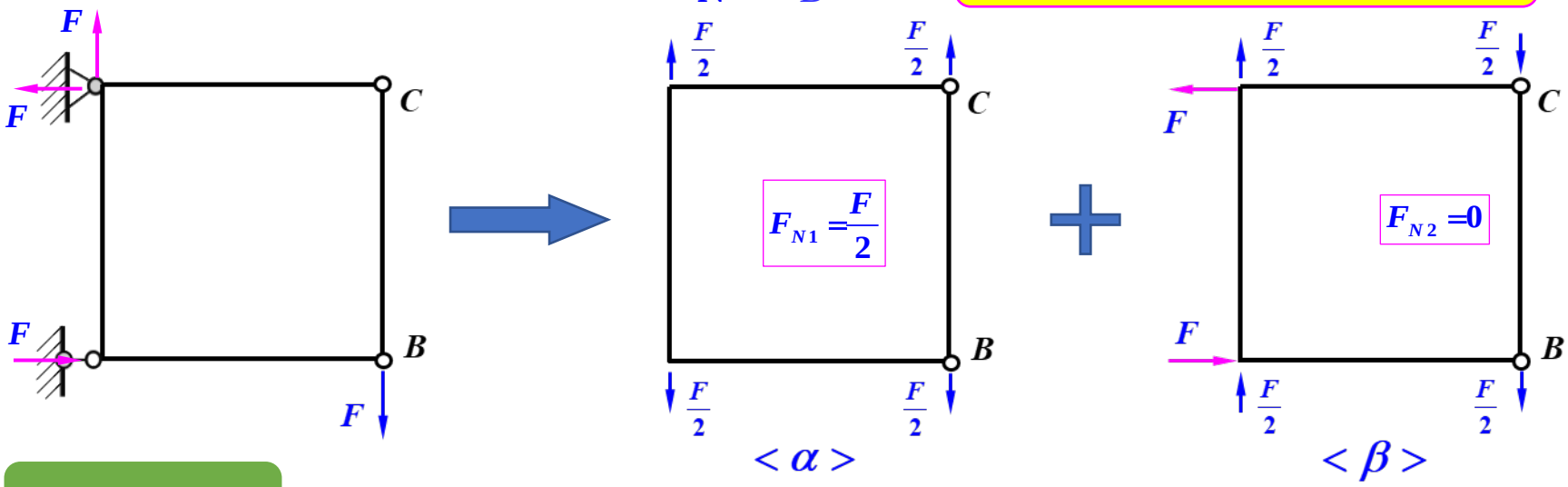




对称性的应用

习题14-7(a) 求BC杆轴力F_N，f_B

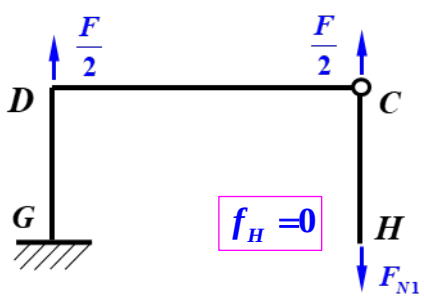
载荷的对称与反对称分解



对称状态

内力按刚度分配

反对称受力



$$f_H=\frac{F_{N1}\frac{a}{2}}{EA}+\frac{(F_{N1}-\frac{F}{2})a^3}{3EI}+\frac{(F_{N1}-\frac{F}{2})a\cdot\frac{a}{2}}{2EI}\cdot a$$

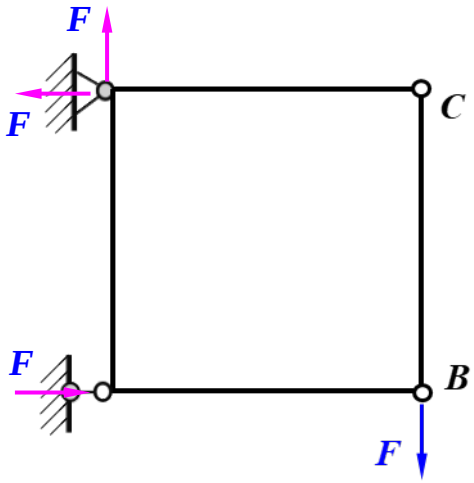
$$F_{N1}=\frac{5Aa^2}{5Aa^2+3I}\frac{F}{2}=\frac{50}{53}\frac{F}{2}$$

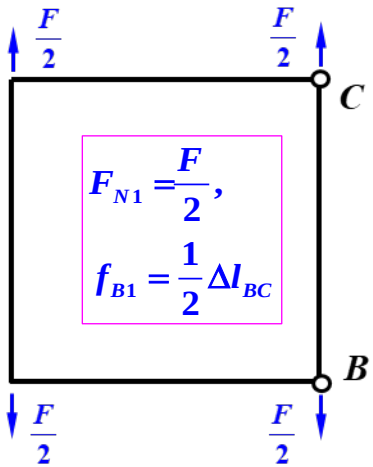
$$f_{B1}=\frac{1}{2}\Delta l_{BC}$$

对称性的应用

习题14-7(a) 求BC杆轴力F_N，f_B

外载荷的对称与反对称分解

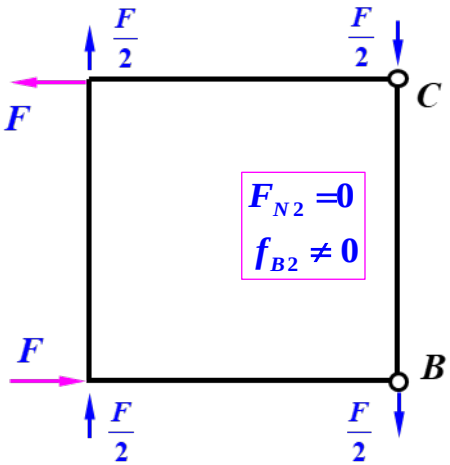




$$F_{N1}=\frac{F}{2},\quad f_{B1}=\frac{1}{2}\Delta l_{BC}$$

$<\alpha>$

内力按刚度分配

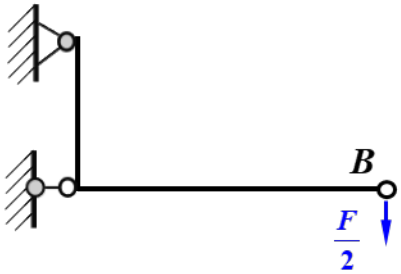


$$F_{N2}=0,\quad f_{B2}\neq 0$$

$<\beta>$

反对称受力

反对称状态

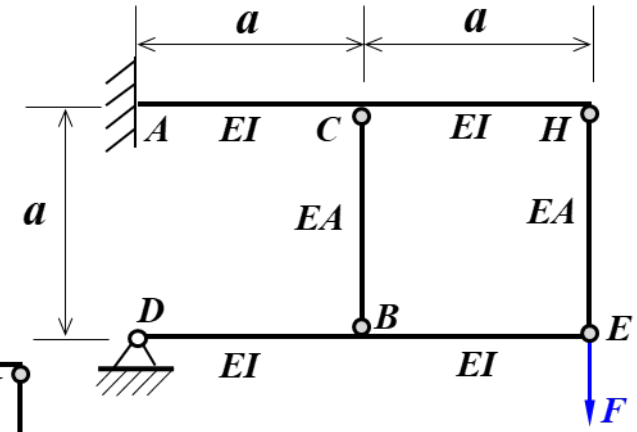


$$f_{B2}=\frac{\frac{F}{2}a^3}{3EI}+\frac{\frac{Fa}{2}\cdot\frac{a}{2}}{3EI}\cdot a$$

$$f_{B2}=\frac{Fa^3}{4EI}$$

选好基本系统

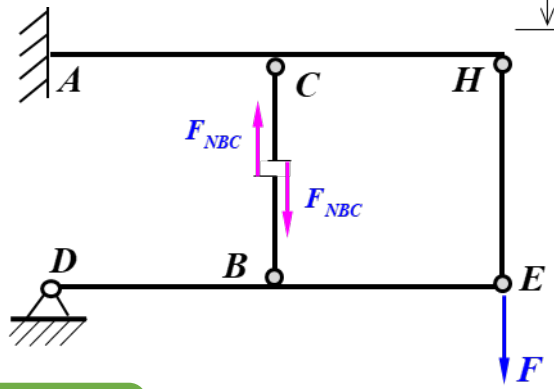
习题14-7(b) 图示梁杆结构，EI为常数，$I=Aa^2/10$。求：F_{NBC}，f_B

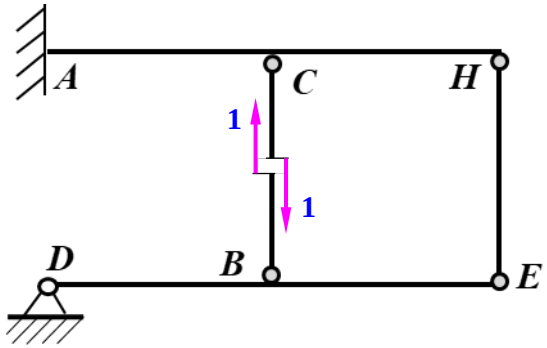


解： 切开BC杆

$$\Delta_{m/m'}=0$$

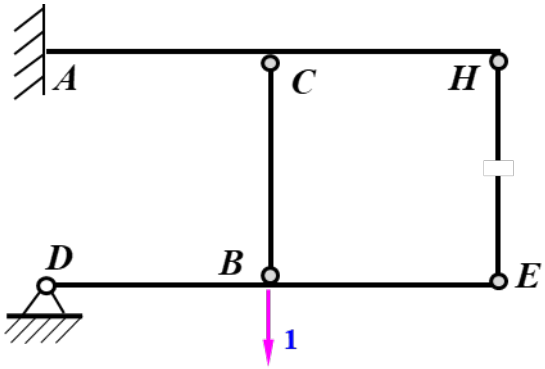
$$F_{NBC}=\frac{6(Aa^2+I)}{4Aa^2+15I}F$$

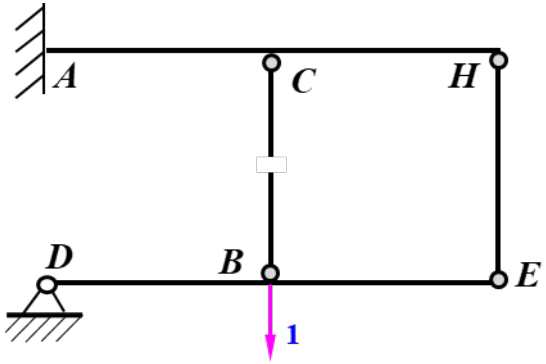




基本系统加单位力求位移

$$f_B=w_C+\Delta l_{BC}$$





- 基本系统不唯一；
- 取最短传力路线。

选好基本系统

习题14-17 图示桁架，*EA*为常数，3杆短 δ，求各杆轴力。

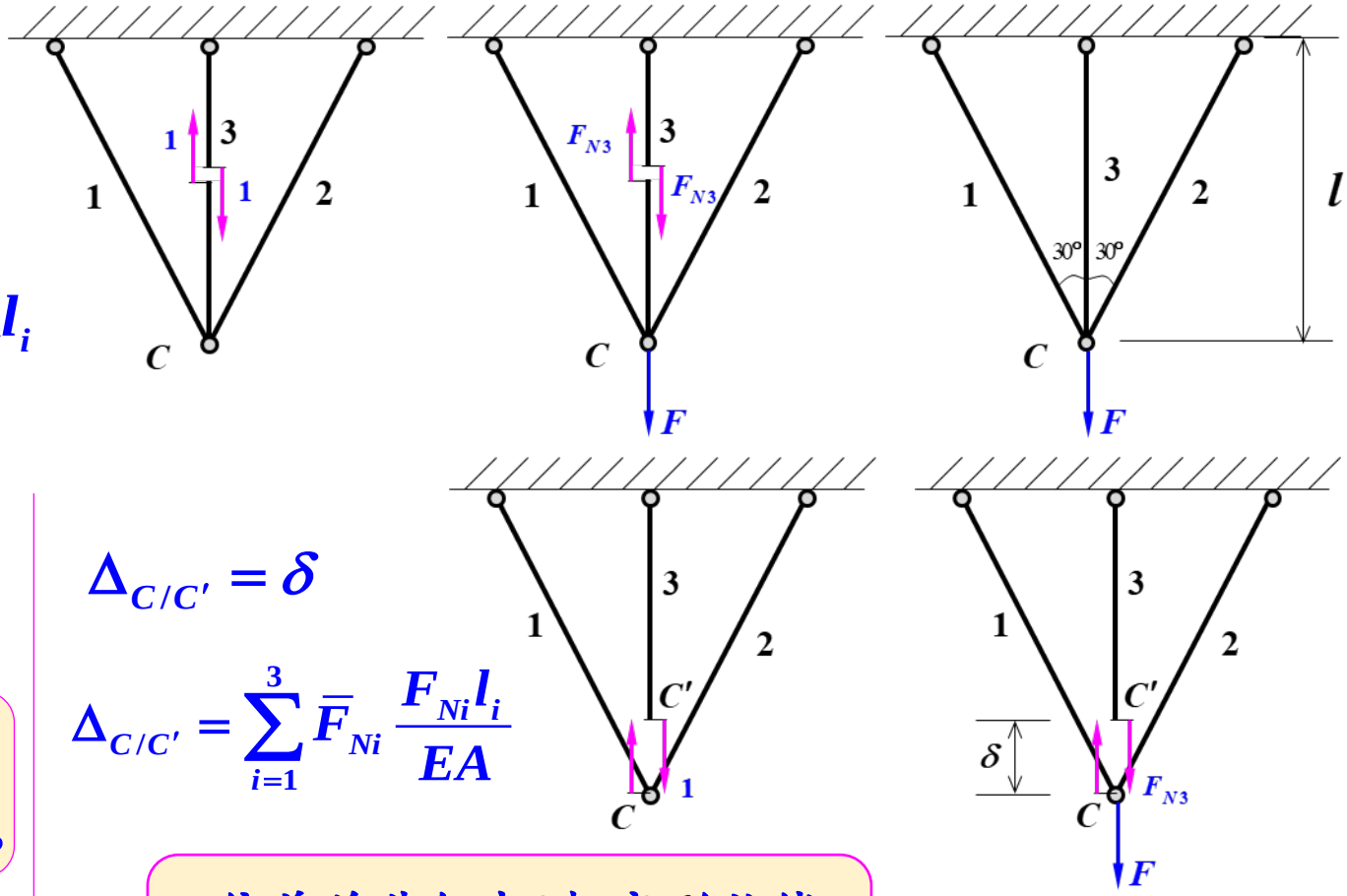
机械、僵化！

$$\Delta_{m/m'} = 0$$

$$\Delta_{m/m'} = \sum_{i=1}^{3} \bar{F}_{Ni} \Delta l_i$$

$$\Delta l_3 = \frac{F_{N3} l}{EA} - \delta$$

轴力引起的变形需“弥补”制造偏差。



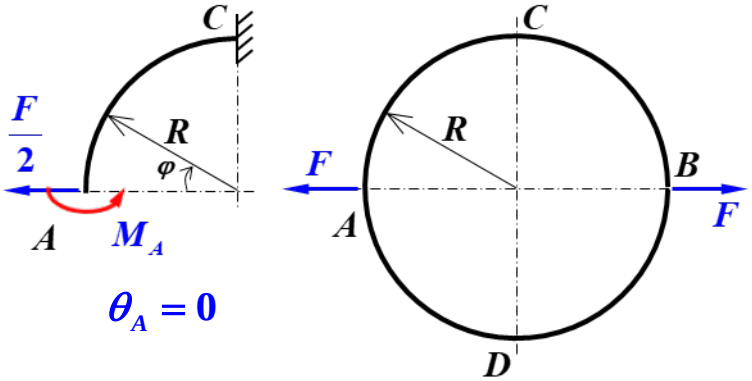
$$\Delta_{C/C'} = \delta$$

$$\Delta_{C/C'} = \sum_{i=1}^{3} \bar{F}_{Ni} \frac{F_{Ni} l_i}{EA}$$

- 偏差并非仅由3杆变形补偿；
- 放到协调条件中更清晰。

静不定结构求解

习题14-11 小曲率圆环，求Δ_{AB}.

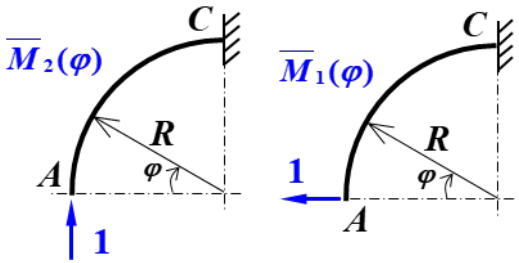


$$M(\varphi)=M_A-\frac{F}{2}R\sin\varphi$$

$$\theta_A=\int_0^{\frac{\pi}{2}}\frac{M(\varphi)}{EI}\frac{\partial M}{\partial\varphi}R\,\mathrm{d}\varphi \qquad M_A=\frac{FR}{\pi}$$

$$\theta_A=0$$

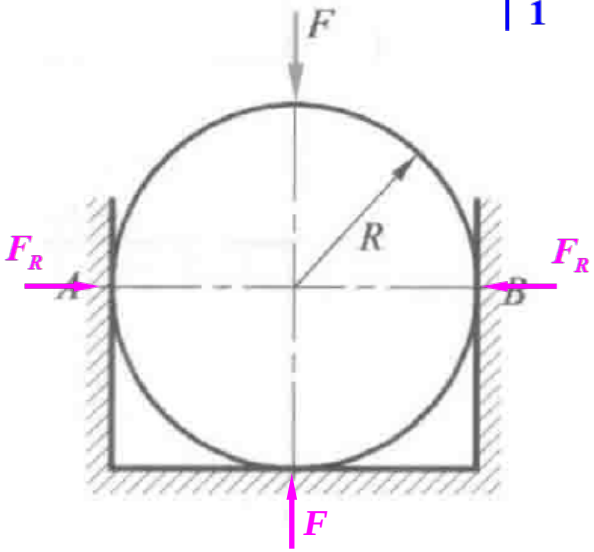
$$\overline{M}_1(\varphi)=-R\sin\varphi \qquad \overline{M}_2(\varphi)=-R(1-\cos\varphi)$$



$$\Delta_{AB}=2\int_0^{\frac{\pi}{2}}\frac{\overline{M}_1M}{EI}R\,\mathrm{d}\varphi \qquad \Delta_{CD}=\frac{4-\pi}{2\pi}\frac{FR^3}{EI}(\downarrow\uparrow)$$

$$=\frac{\pi^2-8}{4\pi}\frac{FR^3}{EI}(\leftarrow\rightarrow)$$

习题14-14 小曲率圆环，求反力



$$\Delta_{AB}=0 \quad\rightarrow\quad F_R$$

静不定结构的冲击

习题15-15 图示小曲率圆环平均半径R，横截面为圆，直径d，材料的弹性模量为E，重物P自由下落，求环内最大应力。

冲击 + 对称圆环三度静不定

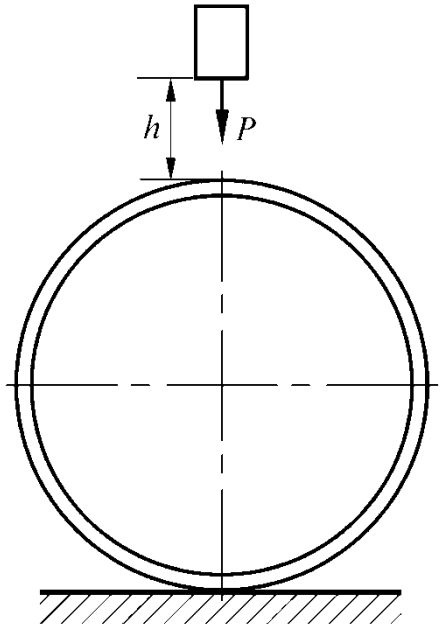
$$(h+f_A)P=V_\varepsilon$$

求解静不定

（习题14-11）

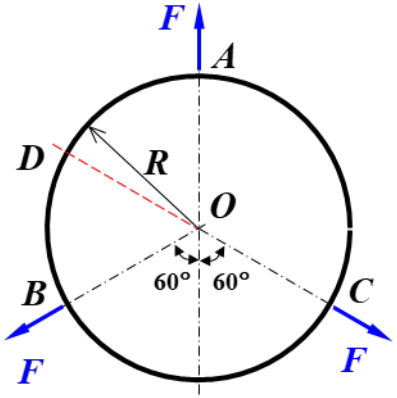
$$P_d \longrightarrow \sigma_{\max}$$

冲击载荷是关键



卡氏定理求广义位移

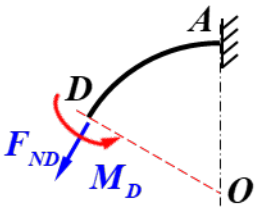
习题14-12 小曲率圆环，求Δ_{AO}.



1）受力与变形中心对称

$$\Delta_{AO} = \frac{1}{3}\frac{\partial V_\varepsilon}{\partial F}$$

2）解静不定圆环



$$\theta_D = 0 \quad \rightarrow \quad M_D$$

反对称内力求解

习题14-14 小曲率圆环，求内力。

$$M_e = 2\pi q R^2$$

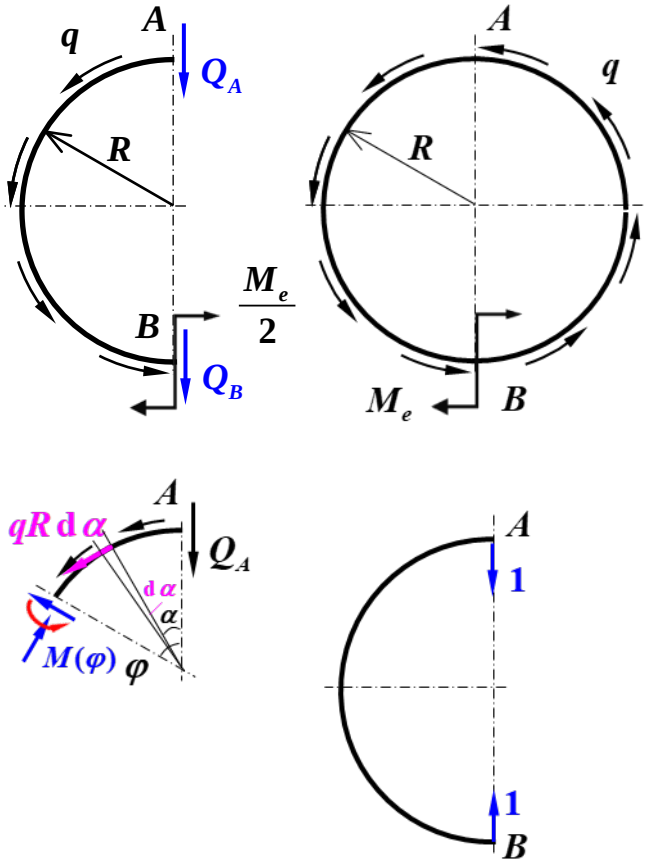
$$Q_A + Q_B + \int_0^{\pi} qR\,\mathrm{d}\varphi \cdot \sin\varphi = 0$$

$$\Delta_{A/B} = 0$$

$$M(\varphi) = Q_A R \sin\varphi - \int_0^{\varphi} qR\,\mathrm{d}\alpha \cdot R\left[1 - \cos(\varphi - \alpha)\right]$$

$$\overline{M}(\varphi) = R\sin\varphi$$

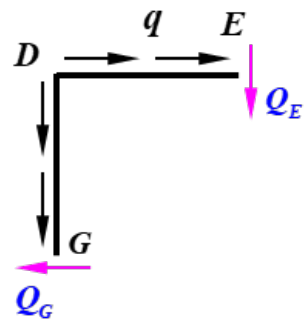
$$Q_A = qR, \quad Q_B = -3qR$$

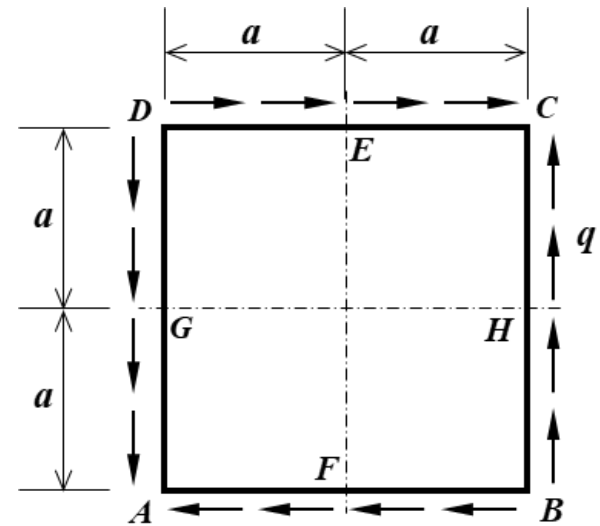


反对称位移求解

例 图示正方形刚架，

EI为常数，求$\Delta_{E/F}$.





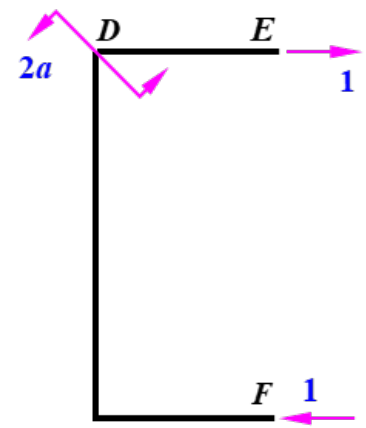
1）求内力

双反对称受力

$Q_E = -qa$， $Q_G = qa$

2）求$\Delta_{E/F}$

单位载荷状态不平衡



结构关于DB对称

$\theta_D = 0$

$$2a \cdot \theta_D + 1 \cdot \Delta_{E/F} = \int_{l_{EDGAF}} \frac{\overline{M}(x) M(x)}{EI} \mathrm{d}x$$

本讲：

下讲：